

Machine Learning Project

Retail Customer Behavioral Analysis

REPORT

Project: Gift E-commerce

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Academic Year: 2025-2026

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1 Introduction

This report presents the work carried out during the Machine Learning workshop on customer behavioral analysis for a gift-focused e-commerce store. The main objective of this project is to develop machine learning solutions that can:

- **Personalize the marketing strategy** through customer segmentation
- **Reduce churn rate** with predictive modeling
- **Increase revenue** by targeting at-risk customers

The provided dataset contains **4,372 customers** described by **52 variables**, including RFM metrics (Recency, Frequency, Monetary), demographic data, purchasing behavior, and account information.

1.1 Project Architecture

A well-structured software architecture is essential for maintainability, reproducibility, and deployment of Machine Learning solutions. This subsection describes the project directory structure.

1.1.1 Directory Structure

The project follows a standard data science layout, clearly separating raw data, processed data, code, models, and outputs.

```
1 ml-retail/
2 |-- data/
3 |   |-- raw/                # Raw data
4 |   |-- processed/          # Cleaned data
5 |   |-- train_test/         # Train/test splits
6 |-- models/                 # Trained models
7 |-- reports/                # Visualizations
8 |-- src/                    # Source code
9 |   |-- preprocessing.py     # Preprocessing pipeline
10 |   |-- train_model.py       # Model training
11 |   |-- segmentation.py      # Customer segmentation
12 |   |-- predict.py           # Prediction script
13 |   |-- utils.py             # Utility functions
```

2 Data Preprocessing

Preprocessing is a critical step to ensure model quality. We implemented a complete pipeline in the file `src/preprocessing.py`.

2.1 Handling Missing Values

Several strategies were used to handle missing values depending on each variable's nature.

2.1.1 Age Variable

The **Age** variable contained **1,311 missing values** (29.99% of the dataset). We used **K-Nearest Neighbors (KNN)** imputation with the following parameters:

- Number of neighbors: $k = 5$
- Inverse-distance weighting
- Variables used: Recency, Frequency, MonetaryTotal, MonetaryAvg, CustomerTenure-Days, FirstPurchaseDaysAgo

This approach captures correlations between age and purchasing behavior, producing more realistic imputations than a simple mean or median.

2.1.2 AvgDaysBetweenPurchases Variable

This variable had **79 missing values**, imputed using the median (1.15 days).

2.2 Outlier and Sentinel Value Processing

Some variables contained **sentinel values** used to encode missing data or errors:

Variable	Anomalous values	Occurrence count
SupportTicketsCount	-1, 999	130 (2.97%)
SatisfactionScore	-1, 0, 99	349 (7.98%)

Table 1: Identified and corrected sentinel values

These values were replaced by each variable's median and then clipped to valid ranges (0-15 for SupportTicketsCount, 1-5 for SatisfactionScore).

2.3 Feature Engineering

To improve predictive performance, we created additional features from raw data.

2.3.1 Registration Date

The **RegistrationDate** variable was parsed and split into four new variables:

- **RegYear**: Registration year
- **RegMonth**: Registration month
- **RegDay**: Registration day
- **RegWeekday**: Day of week (0=Monday, 6=Sunday)

2.3.2 IP Address

From the **LastLoginIP** variable, we extracted:

- **IsPrivateIP**: Boolean flag (326 customers with private IPs)
- **LoginCountry**: Login country (using the DB-IP Lite database)

The login country distribution shows a mostly US customer base (1,494 customers), followed by China (362) and Japan (219).

2.4 Removing Unreliable Variables

Some variables were excluded from preprocessing due to questionable reliability.

The variable `NewsletterSubscribed` was removed because it was marked as unreliable in the project documentation.

3 Data Leakage Detection and Correction

A detailed analysis of correlations between features and the `Churn` target revealed several cases of **data leakage**, which would have compromised model validity.

3.1 Excluded Variables

The following table summarizes the removed variables and why they were excluded.

Variable	Reason for exclusion
<code>ChurnRiskCategory</code>	Directly derived from the target variable (100% correlation)
<code>RFMSegment</code>	The "Dormants" category corresponds to 100% churned customers
<code>CustomerType</code>	The "Lost" category corresponds to 100% churned customers
<code>Recency</code>	Perfect split: $\text{Churn}=0$ if $\text{Recency} \leq 90$ days, $\text{Churn}=1$ if $\text{Recency} > 90$ days

Table 2: Variables removed due to data leakage

3.2 Recency Variable Analysis

The analysis showed that `Recency` itself defines churn:

- Non-churned customers: $\text{Recency} \in [1, 90]$ days
- Churned customers: $\text{Recency} \in [91, 374]$ days

This finding is critical: churn is effectively defined as inactivity for more than 90 days. Using this feature would amount to predicting the past with the past.

4 Predictive Churn Modeling

This section details the process used to build and evaluate classification models for churn prediction. Several approaches were implemented, from preprocessing to comparison of seven different algorithms.

4.1 Data Preparation for Modeling

Before training the models, we applied several essential data transformations.

4.1.1 Feature Encoding

- **Numeric variables (35):** Standardized using `StandardScaler`
- **Categorical variables (14):** One-hot encoded using `OneHotEncoder`

4.1.2 Train/Test Split

The dataset was split into:

- **Training set:** 3,497 samples (80%)
- **Test set:** 875 samples (20%)

Stratification was applied to preserve target distribution (33.26% churn) in both sets.

4.1.3 Class Imbalance Handling

A churn rate of 33.26% represents moderate class imbalance. We applied **SMOTE** (Synthetic Minority Over-sampling Technique) on the training set:

- Before SMOTE: 2,334 non-churned, 1,163 churned
- After SMOTE: 2,334 non-churned, 2,334 churned

4.2 Evaluated Models

Seven classification algorithms were trained and compared:

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost (Tuned)	99.1%	99.0%	98.3%	98.6%	99.97%
Logistic Regression	98.4%	98.6%	96.6%	97.6%	99.8%
Gradient Boosting	98.3%	96.9%	97.9%	97.4%	99.9%
Decision Tree	97.3%	94.9%	96.9%	95.9%	97.2%
SVM	97.1%	98.9%	92.4%	95.6%	99.7%
Random Forest	94.9%	96.6%	87.6%	91.9%	98.7%
KNN	84.1%	70.9%	88.7%	78.8%	92.5%

Table 3: Model performance comparison on the test set

4.3 Hyperparameter Optimization

To further improve the best-performing model, XGBoost (selected based on F1 score), we performed hyperparameter tuning with **GridSearchCV** and 5-fold cross-validation.

Optimal Hyperparameters - XGBoost

- `learning_rate`: 0.1
- `max_depth`: 7
- `n_estimators`: 100
- `subsample`: 0.8

Cross-validation F1 score: 99.09%

4.4 Feature Importance

Understanding which factors drive model predictions is essential for interpretability and actionability. XGBoost feature importance highlights the strongest churn predictors:

1. **LoyaltyLevel_Établi** (35.3%): Established customers are significantly less likely to churn
2. **FavoriteSeason_Hiver** (20.4%): Seasonal buying patterns are highly predictive
3. **PreferredMonth** (20.0%): Preferred purchase month indicates engagement level
4. **LoyaltyLevel_Nouveau** (5.8%): New customers show distinct behavioral patterns
5. **CustomerTenureDays** (4.5%): Customer tenure

5 Customer Segmentation

In addition to churn prediction, a complementary customer segmentation approach was developed. This RFM analysis supports personalized marketing strategies and helps identify customer groups with distinct characteristics and needs.

5.1 Methodology

Segmentation was performed using a quantitative approach based on the K-Means algorithm.

5.1.1 RFM Variables

- **Recency**: Number of days since the last purchase
- **Frequency**: Total number of purchases
- **Monetary**: Total amount spent

5.1.2 Clustering Algorithm

We used **K-Means** with the following steps:

1. Standardization of RFM variables using **StandardScaler**
2. Search for the optimal number of clusters using silhouette score
3. Application of K-Means with $k = 3$ clusters

5.2 Identified Segments

Running K-Means revealed three distinct segments with very different RFM profiles and churn rates.

Segment	Customers	Churn Rate	Avg. Recency	Avg. Freq.	Avg. Amount
At Risk	1,109 (25.4%)	100%	246 days	1.85	460 \$
Champions	26 (0.6%)	0%	6 days	83.35	75,966 \$
Active Customers	3,237 (74.0%)	11%	40 days	5.55	1,796 \$

Table 4: Profile of identified customer segments

5.3 Segment-Based Marketing Recommendations

To maximize the effectiveness of marketing interventions, specific recommendations were defined for each segment.

Champions Segment (Priority: HIGH VALUE)

Recommended actions:

- Exclusive offers and early access to new products
- Loyalty program with VIP benefits
- Review requests and referral program
- Personalized thank-you communications

Channels: Email, Direct mail, Dedicated account manager

At-Risk Segment (Priority: URGENT)

Recommended actions:

- Win-back campaign with special discounts
- Survey to understand inactivity drivers
- Personalized "We miss you" messages
- Time-limited exclusive offers

Channels: Email, SMS, Direct mail

Active Customers Segment (Priority: MEDIUM)

Recommended actions:

- Upselling toward higher-value products
- Cross-selling complementary products
- Free-shipping thresholds to increase basket size
- Early access to seasonal sales

Channels: Email, Retargeting ads

6 Generated Visualizations

The project produces six key visualizations stored in the `reports/` directory. Each chart provides critical insights into model performance and customer behavior, covering model comparison, predictive evaluation, and segment analysis.

6.1 Model Comparison

This visualization includes two side-by-side bar charts for fast comparison of the seven algorithms:

- **Left panel:** Classification metrics (Accuracy, Precision, Recall, F1) for each model
- **Right panel:** ROC-AUC score per model

It quickly identifies XGBoost as the top-performing model across all criteria.

6.2 ROC Curves

Beyond core metrics, a deeper view of each model's discrimination capability is needed. ROC curves display all models on the same plot:

- X-axis: False Positive Rate (FPR)
- Y-axis: True Positive Rate (TPR)
- The diagonal line represents a random classifier ($AUC = 0.5$)

Curves closer to the top-left corner indicate better performance.

6.3 Confusion Matrix

For a granular understanding of classification errors made by the best model, a detailed confusion matrix is generated. It presents XGBoost predictions as a heatmap:

- **True Negatives (TN):** Customers correctly identified as non-churned
- **True Positives (TP):** Customers correctly identified as churned
- **False Positives (FP):** Non-churned customers incorrectly flagged (Type I error)
- **False Negatives (FN):** Churned customers missed by the model (Type II error)

6.4 Feature Importance

To understand churn drivers, feature importance analysis is essential. A horizontal bar chart shows the top 20 most influential XGBoost features, ranked by their contribution to reducing prediction error.

6.5 Segment Distribution

Unlike classification metrics, this visualization focuses on behavioral patterns discovered through segmentation. It presents clustering outcomes from two perspectives:

- **Pie chart:** Customer distribution by segment
- **Bar chart:** Churn rate by segment

6.6 RFM Profile Heatmap

To summarize the distinctive characteristics of each segment, a normalized heatmap provides an overview of RFM profiles. It displays a color matrix of normalized RFM values by segment:

- Rows: RFM metrics (Recency, Frequency, Monetary)
- Columns: Customer segments
- Values: Normalized scores between 0 and 1

7 Flask Interface Deployment

To complete the deployment phase of the project, we implemented a Flask-based interface that operationalizes the trained churn model and segmentation outputs for end users.

7.1 Deployment Objective

The objective of this interface is to transform the analytical pipeline into a usable application where a user can:

- Predict churn for a specific customer
- Score customers in batch mode from the processed dataset
- Interpret results through risk levels, segment assignment, and recommended marketing actions

7.2 Implemented Application Structure

The deployment layer was organized in the `src/app/` package:

- `web.py`: Flask routes and request handling
- `services.py`: model loading, inference logic, segmentation enrichment, and summaries
- `templates/`: HTML templates (`base.html`, `index.html`)
- `static/style.css`: interface styling

This structure separates presentation, routing, and business logic to improve maintainability.

7.3 Functionalities Delivered

7.3.1 Single-Customer Prediction

The interface accepts a `CustomerID`, retrieves the corresponding record from the processed dataset, and displays:

- Churn probability and churn decision
- Risk level (Low, Medium, High, Critical)
- Segment assignment inferred from RFM values
- Segment-specific marketing recommendations and channels

7.3.2 Batch Processing with Controlled Pagination

The *Process all customers* workflow was optimized for responsiveness:

- Only the first 100 customers from `data/processed/retail_customers_cleaned.csv` are scored
- Results are sorted by ascending churn priority: Low → Medium → High → Critical
- Pagination is applied at 10 rows per page with previous/next navigation

7.3.3 Operational API Endpoints

In addition to the web interface, lightweight endpoints were exposed:

- `/health`: service health check
- `/api/summary`: model and dataset summary
- `/api/predict`: JSON prediction endpoint

7.4 User Interface Adjustments

To align the interface with a more professional presentation, the UI was refined with:

- Header removal for a cleaner layout
- Neutral grayscale palette replacing the previous blue accent
- Square/block visual style (no rounded corners)

These adjustments improved readability and consistency for report/demo usage.

8 Conclusion

This project delivered a complete customer behavioral analytics solution including:

1. **A robust preprocessing pipeline** handling missing values (KNN for age), anomalous values (sentinel values), and feature engineering (temporal and geographic feature extraction).
2. **Rigorous data leakage detection** that led to excluding 4 variables (ChurnRiskCategory, RFMSegment, CustomerType, Recency) that would have artificially inflated model performance.
3. **A high-performing churn prediction model** (XGBoost) achieving an F1 score of 98.6% and ROC-AUC of 99.97% on the test set.
4. **Actionable customer segmentation** identifying three distinct segments with tailored marketing recommendations.

8.1 Future Work

Several improvement paths can be considered:

- Integrating temporal data for time-series analysis
- Building an interactive dashboard for real-time monitoring
- Deploying the model through a REST API
- Running A/B tests on segment-based marketing recommendations